

The Type-I Hypergraph Hanner Conjecture

Candidate full solution for arXiv:0903.3237

Model: GPT5.6

19 July 2026

Status: full solution, likely valid. Every semi-norming Type-I hypergraph pair with parameter $s \geq 2$ satisfies the conjectured $|H|$ -Hanner inequality. This proves Conjecture 3.15 of Hatami [1]; non-factorizability is not needed.

1 Source problem

Let $H = (\alpha, \beta)$ be a finite hypergraph pair. In Hatami's notation, H is of Type I with parameter s if, on its support $\mathcal{S}$,

$$\alpha(\omega) = \beta(\omega) = \frac{s}{2}.$$

If $m = |\mathcal{S}|$, then $|H| = ms$. For a function f on the underlying product measure space, the associated hypergraph norm satisfies

$$\|f\|_H^{ms} = \int \prod_{\omega \in \mathcal{S}} |f(\omega(x))|^s dx. \quad (1)$$

Conjecture 3.15 asks whether, for $s \geq 2$,

$$\|f + g\|_H^{|H|} + \|f - g\|_H^{|H|} \leq (\|f\|_H + \|g\|_H)^{|H|} + \left| \|f\|_H - \|g\|_H \right|^{|H|}. \quad (2)$$

Conjecture 3.15 *Let $H = (\alpha, \beta)$ be a non-factorizable semi-norming hypergraph pair of Type I with parameter $s \geq 2$. Then every L_H space satisfies the $|H|$ -Hanner inequality.*

Figure 1: Conjecture 3.15 on page 25 of the source paper.

The source proves the corresponding inequality on ℓ_H when s is an even integer by expanding the s th powers directly. The point below is to perform only the finite expansion indexed by the m support elements, leaving the real exponent s inside an ordinary L_s norm; the argument works on an arbitrary underlying measure space.

2 Two elementary ingredients

We use Hatami's generalized Hölder inequality [1, Lemma 2.10]. In the special form needed here, if $A \subseteq \mathcal{S}$ and $a = \|f\|_H$, $b = \|g\|_H$, then

$$\int \prod_{\omega \notin A} |f(\omega(x))|^s \prod_{\omega \in A} |g(\omega(x))|^s dx \leq a^{s(m-|A|)} b^{s|A|}. \quad (3)$$

Indeed, decompose H into the two nonnegative hypergraph pairs supported on $\mathcal{S} \setminus A$ and A , and apply the nonnegative-function clause of that lemma to $|f|$ and $|g|$.

We will also need the following small monotonicity observation.

Lemma 1. *For $s \geq 1$, the function*

$$\Phi_s(u, v) = (u + v)^s + |u - v|^s$$

is nondecreasing in each variable on $[0, \infty)^2$.

Proof. On the region $u \geq v$, its partial derivatives are

$$s((u + v)^{s-1} + (u - v)^{s-1}), \quad s((u + v)^{s-1} - (u - v)^{s-1}),$$

and both are nonnegative. The other region follows by symmetry. $\square$

3 Idea of the proof

The obstruction in the source proof is the temptation to expand the real s th power, which is possible only when s is an even integer. Instead we factor the Type-I integrand as the ordinary L_s norm of the product over the m support elements. That product has an integer-degree expansion for every real s : its even and odd terms form two functions E and O . The ordinary L_s Hanner inequality controls $E + O$ and $E - O$, while Hatami's generalized Hölder lemma controls each mixed monomial in E and O . The two resulting bounds are precisely the even and odd binomial sums; coordinatewise monotonicity of the two-variable Hanner expression then recombines them into the conjectured right-hand side.

4 Full solution

Theorem 2. *Let H be a semi-norming hypergraph pair of Type I with parameter $s \geq 2$. Then every L_H space satisfies the $|H|$ -Hanner inequality (2). In particular, Conjecture 3.15 of [1] holds.*

Proof. Fix $f, g \in L_H$, and abbreviate $a = \|f\|_H$, $b = \|g\|_H$. For $A \subseteq \mathcal{S}$, put

$$M_A(x) = \prod_{\omega \notin A} f(\omega(x)) \prod_{\omega \in A} g(\omega(x)).$$

Separate the expansion into its even and odd parts:

$$E = \sum_{|A| \text{ even}} M_A, \quad O = \sum_{|A| \text{ odd}} M_A.$$

Ordinary distributivity gives

$$E + O = \prod_{\omega \in \mathcal{S}} (f(\omega(x)) + g(\omega(x))), \quad E - O = \prod_{\omega \in \mathcal{S}} (f(\omega(x)) - g(\omega(x))). \quad (4)$$

Apply the classical Hanner inequality in the ordinary complex L_s space [2]:

$$\|E + O\|_s^s + \|E - O\|_s^s \leq \Phi_s(\|E\|_s, \|O\|_s). \quad (5)$$

By (3),

$$\|M_A\|_s \leq a^{m-|A|}b^{|A|}.$$

The triangle inequality in L_s therefore yields

$$\begin{aligned} \|E\|_s &\leq E_0 := \sum_{k \text{ even}} \binom{m}{k} a^{m-k} b^k, \\ \|O\|_s &\leq O_0 := \sum_{k \text{ odd}} \binom{m}{k} a^{m-k} b^k. \end{aligned}$$

Lemma 1 permits these two upper bounds to be substituted simultaneously into (5). The even and odd binomial identities give

$$E_0 + O_0 = (a + b)^m, \quad E_0 - O_0 = (a - b)^m.$$

Consequently,

$$\|E + O\|_s^s + \|E - O\|_s^s \leq (a + b)^{ms} + |a - b|^{ms}. \quad (6)$$

Finally, (1) and (4) identify the left-hand side of (6) with $\|f + g\|_H^{ms} + \|f - g\|_H^{ms}$. Since $ms = |H|$, this is exactly (2). $\square$

5 Checks and scope

1. The proof never expands a noninteger power. It expands a product with integer degree m and then invokes the classical L_s inequality.
2. Complex-valued functions cause no change: (4) is algebraic, the mixed Hölder estimate is applied to absolute values, and the classical Hanner inequality holds in complex L_s .
3. Non-factorizability is absent from the argument. Thus the theorem is slightly stronger than the source conjecture.
4. The restriction $s \geq 2$ occurs only in the direction of the ordinary L_s Hanner inequality. For $s < 2$ its direction reverses, consistent with the obstruction recorded immediately before Conjecture 3.15.

For a numerical diagnostic, the accompanying script tests the first non-even example: the Type-I norm generated by the four-cycle with $s = 3$. Random search and differential evolution found no positive Hanner defect; the optimized value was within $3.1 \cdot 10^{-15}$ of equality. This check is not used in the proof.

6 Novelty note

The run’s registry, solution, attempt, and proof-gap indexes contained no entry for arXiv:0903.3237. Exact searches for the source conjecture, “Type-I hypergraph pair,” and “Hanner inequality” found Hatami’s paper and thesis, but no later proof. Nearby literature on complex L_p Hanner inequalities and Schatten norms does not state the even/odd hypergraph argument above. This was a bounded search, not a complete citation audit.

References

- [1] H. Hatami, *On generalizations of Gowers norms and their geometry*, arXiv:0903.3237 (2009), especially Lemma 2.10 and Conjecture 3.15.
- [2] O. Hanner, *On the uniform convexity of L^p and ℓ^p* , Arkiv för Matematik 3 (1956), 239–244.